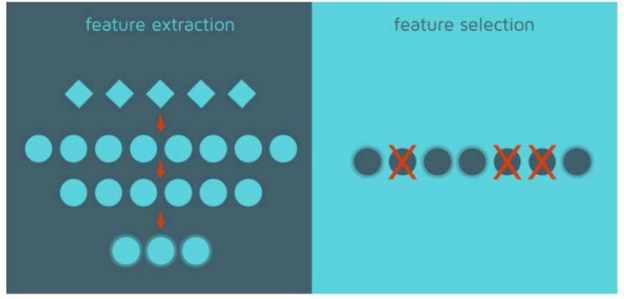
Dimensionality Reduction



01 **Feature Selection**

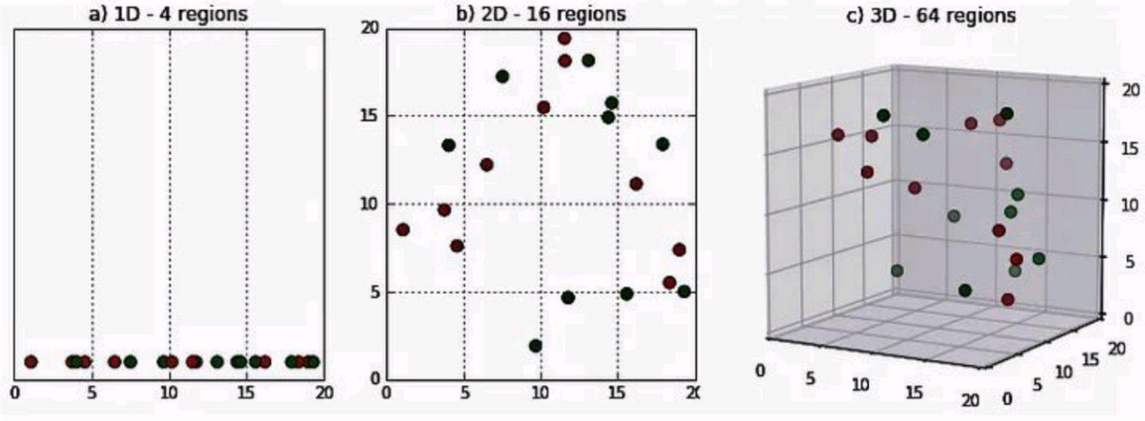
Original features are maintained

Keep most important features

02 **Feature Extraction**

Features are transformed to a new space

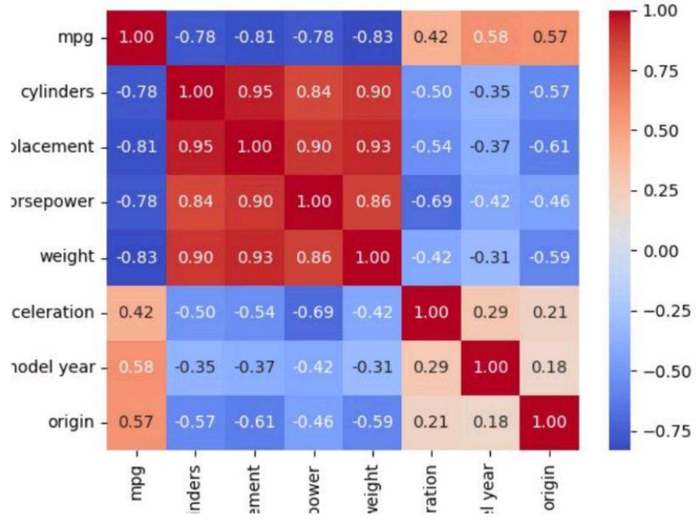
Curse of Dimensionality



Feature Selection



Feature Selection - Correlation Coefficient



Feature Selection - Variance Threshold

```python
import pandas as pd
from sklearn.feature_selection import VarianceThreshold

data = pd.read_csv("diabetes.csv", usecols=[i for i in range(8)])
variance_threshold = VarianceThreshold(threshold=1)
variance_threshold.fit(data)
print(variance_threshold.variances_)
print(variance_threshold.get_support())
```

```
fuel_viz
C:\Users\Viet\anaconda3\envs\basic\python.exe C:\Users\Viet\Pycharm
[1.13392724e+01 1.02091726e+03 3.74159449e+02 2.54141900e+02
 1.32638869e+04 6.20790465e+01 1.09635697e-01 1.38122964e+02]
[ True  True  True  True  True  True False  True]
```

Feature Selection - Lasso (L1)

```python
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline

data = pd.read_csv("diabetes.csv")
x = data.drop("Outcome", axis=1)
y = data["Outcome"]
cls = Pipeline(steps=[
    ('scaler', StandardScaler()),
    ('model', LogisticRegression(penalty="l1", solver="liblinear"))
])
cls.fit(x, y)
for i, j in zip(x.columns, cls["model"].coef_[0]):
    print("{} is with coefficient of {}".format(i, j))
```

```
C:\Users\Viet\anaconda3\envs\basic\python.exe C:\Users\Viet\PycharmP
Pregnancies is with coefficient of 0.4047757715484604
Glucose is with coefficient of 1.1020534150332097
BloodPressure is with coefficient of -0.23905878835657515
SkinThickness is with coefficient of 0.0
Insulin is with coefficient of -0.1201412360687196
BMI is with coefficient of 0.6883338771775394
DiabetesPedigreeFunction is with coefficient of 0.30157785094254597
Age is with coefficient of 0.16766878675897076
```

```python
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline

data = pd.read_csv("diabetes.csv")
x = data.drop("Outcome", axis=1)
y = data["Outcome"]
cls = Pipeline(steps=[
    ('scaler', StandardScaler()),
    ('model', LogisticRegression(penalty="l2", solver="lbfgs"))
])
cls.fit(x, y)
for i, j in zip(x.columns, cls["model"].coef_[0]):
    print("{} is with coefficient of {}".format(i, j))
```

```
C:\Users\Viet\anaconda3\envs\basic\python.exe C:\Users\Viet\PycharmF
Pregnancies is with coefficient of 0.4086468712956544
Glucose is with coefficient of 1.107111971571363
BloodPressure is with coefficient of -0.2508680388517632
SkinThickness is with coefficient of 0.00905045508333392
Insulin is with coefficient of -0.13083627394872085
BMI is with coefficient of 0.6963087173103848
DiabetesPedigreeFunction is with coefficient of 0.3088372414180195
Age is with coefficient of 0.1764978217442397
```

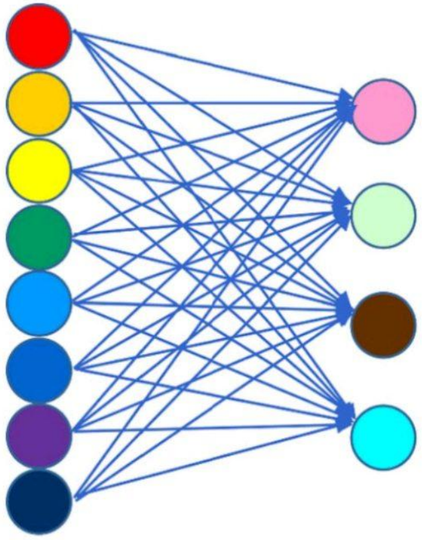
Feature Selection - RandomForest

```python
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline

data = pd.read_csv("diabetes.csv")
x = data.drop("Outcome", axis=1)
y = data["Outcome"]
cls = Pipeline(steps=[
    ('scaler', StandardScaler()),
    ('model', RandomForestClassifier())
])
cls.fit(x, y)
print(cls["model"].feature_importances_)
```

```
fuel_viz
C:\Users\Viet\anaconda3\envs\basic\python.exe C:\Users\Viet\Pycharm
[0.08485069 0.26544363 0.09088871 0.07260613 0.07047188 0.15482689
 0.12826565 0.13264643]
```

Feature Extraction



Feature Extraction - Principle Component Analyst

